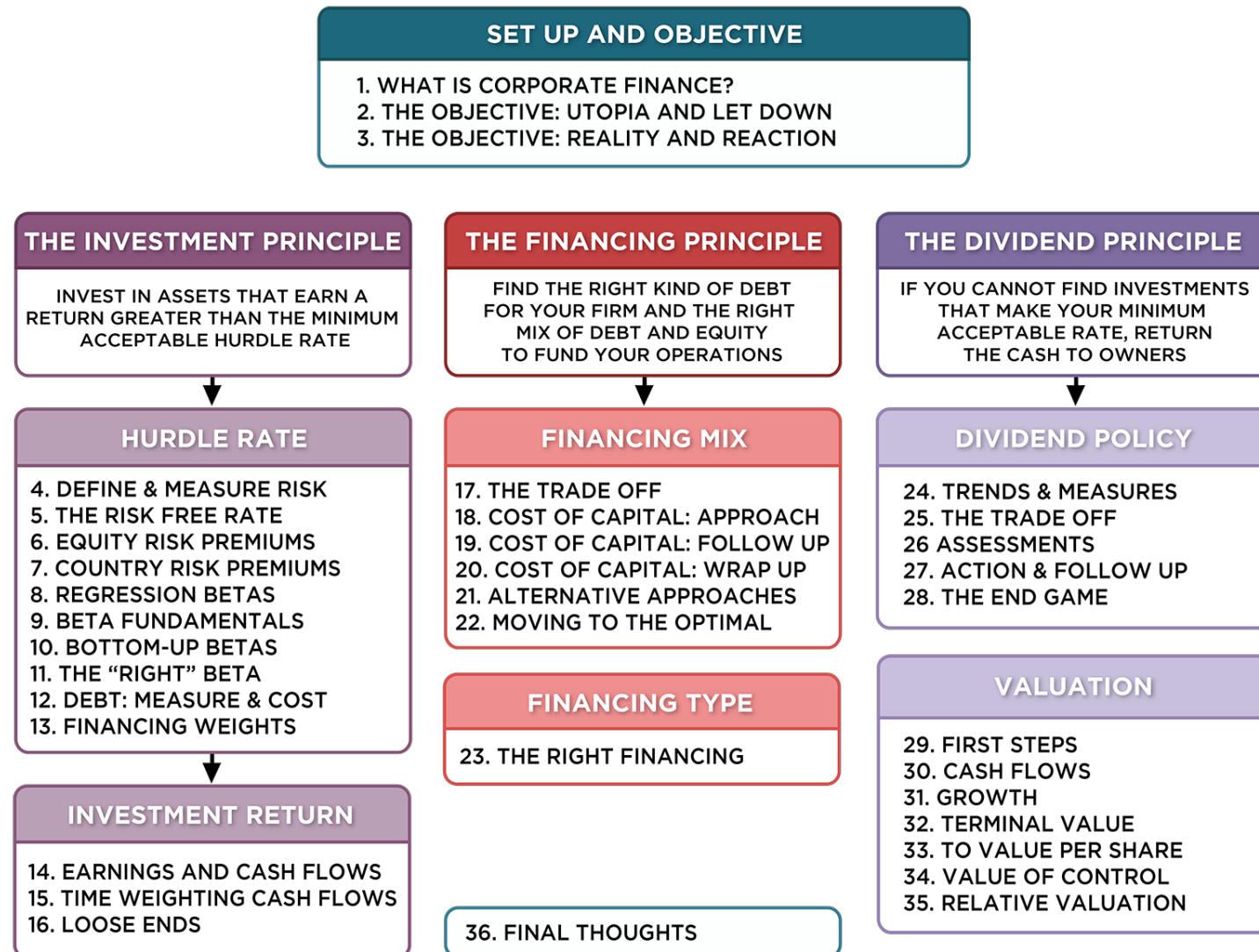


SESSION 12: DEBT: MEASURE AND COST

Debt, the double-edged sword!

SESSION 12: DEBT: MEASURE AND COST



FROM COST OF EQUITY TO COST OF CAPITAL

- The cost of capital is a composite cost to the firm of raising financing to fund its projects.
- In addition to equity, firms can raise capital from debt.

WHAT IS DEBT?

- General Rule: Debt generally has the following characteristics:
 - Commitment to make fixed payments in the future
 - Fixed payments are tax deductible
 - Failure to make the payments can lead to either default or loss of control of the firm to the party to whom payments are due
- As a consequence, debt should include
 - Any interest-bearing liability, whether short term or long term
 - Any lease obligation, whether operating or capital

ESTIMATING THE COST OF DEBT

- If the firm has bonds outstanding, and the bonds are traded, the yield to maturity on a long-term, straight (no special features) bond can be used as the interest rate.
- If the firm is rated, use the rating and a typical default spread on bonds with that rating to estimate the cost of debt.
- If the firm is not rated,
 - and it has recently borrowed long term from a bank, use the interest rate on the borrowing, or
 - estimate synthetic rating for the company, and use the synthetic rating to arrive at a default spread and a cost of debt.
- The cost of debt has to be estimated in the same currency as the cost of equity and the cash flows in the valuation.

THE EASY ROUTE: OUTSOURCING THE MEASUREMENT OF DEFAULT RISK

- For those firms that have bond ratings from global rating agencies, I used those ratings:

Company	S&P Rating	Risk-Free Rate	Default Spread	Cost of Debt
Disney	A	2.75% (US \$)	1.00%	3.75%
Deutsche Bank	A	1.75% (€)	1.00%	2.75%
Vale	A-	2.75% (US \$)	1.30%	4.05%

- If you want to estimate Vale's cost of debt in \$R terms, we can again use the differential inflation approach we used for the cost of equity:

$$\text{COST OF DEBT}_{Rs} = (1 + \$ \text{COST OF DEBT}_{US \$}) \frac{(1 + \text{EXPECTED INFLATION}_{Rs})}{(1 + \text{EXPECTED INFLATION}_{US \$})} - 1$$

$$= (1.0405) \frac{(1.09)}{(1.02)} - 1 = 11.19\%$$

A MORE GENERAL ROUTE: ESTIMATING SYNTHETIC RATINGS

- The rating for a firm can be estimated using the financial characteristics of the firm. In its simplest form, we can use just the interest coverage ratio:

$$\text{INTEREST COVERAGE RATIO} = \text{EBIT} / \text{INTEREST EXPENSES}$$

- For the four non-financial service companies, we obtain the following:

COMPANY	OPERATING INCOME	INTEREST EXPENSES	INTEREST COVERAGE RATIO
DISNEY	\$10,023	\$444	22.57
VALE	\$15,667	\$1,342	11.67
TATA MOTORS	Rs 166,605	Rs 36,972	4.51
BAIDU	Cy 11,193	Cy 472	23.72
BOOKSCAPE	\$2,536	\$492	5.16

INTEREST COVERAGE RATIOS, RATINGS AND DEFAULT SPREADS – NOVEMBER 2013

<i>Large cap (>\$5 billion)</i>	<i>Small cap or risky (<\$5 billion)</i>	<i>Rating is (S&P/ Moody's)</i>	<i>Spread (11/13)</i>
>8.50	>12.5	Aaa/AAA	0.40%
6.5-8.5	9.5-12.5	Aa2/AA	0.70%
5.5-6.5	7.5-9.5	A1/A+	0.85%
4.25-5.5	6-7.5	A2/A	1.00%
3-4.25	4.5-6	A3/A-	1.30%
2.5-3	4-4.5	Baa2/BBB	2.00%
2.25-2.5	3.5-4	Ba1/BB+	3.00%
2-2.25	3-3.5	Ba2/BB	4.00%
1.75-2.25	2.5-3	B1/B+	5.50%
1.5-1.75	2-2.5	B2/B	6.50%
1.25-1.5	1.5-2	B3/B-	7.25%
0.8-1.25	1.25-1.5	Caa/CCC	8.75%
0.65-0.8	0.8-1.25	Ca2/CC	9.50%
0.2-0.65	0.5-0.8	C2/C	10.50%
<0.2	<0.5	D2/D	12.00%

INTEREST COVERAGE RATIOS - NOVEMBER 2013			
DISNEY	22.57	➡	AAA
VALE	11.67	➡	AA
TATA MOTORS	4.51	➡	A-
BAIDU	23.72	➡	AAA
BOOKSCAPE	5.16	➡	A-

SYNTHETIC VERSUS ACTUAL RATINGS: RATED FIRMS

- Disney's synthetic rating is AAA whereas its actual rating is A. The difference can be attributed to any of the following:
 - Synthetic ratings reflect only the interest coverage ratio whereas actual ratings incorporate all of the other ratios and qualitative factors
 - Synthetic ratings do not allow for sector-wide biases in ratings
 - Synthetic rating was based on 2013 operating income whereas actual rating reflects normalized earnings
- Vale's synthetic rating is AA, but the actual rating for dollar debt is A-. The biggest factor behind the difference is the presence of country risk, since Vale is probably being rated lower for being a Brazil-based corporation.
- Deutsche Bank had an A rating. We will not try to estimate a synthetic rating for the bank. Defining interest expenses on debt for a bank is difficult . . .

ESTIMATING COST OF DEBT

- For Bookscape, we will use the synthetic rating (A-) to estimate the cost of debt:
 - Default Spread based upon A- rating = 1.30%
 - Pre-tax cost of debt = Riskfree Rate + Default Spread = 2.75% + 1.30 % = 4.05%
 - After-tax cost of debt + Pre-tax cost of debt (1 – tax rate) = 4.05% (1-0.40) = 2.43%
- For the three publicly traded firms that are rated in our sample, we will use the actual bond ratings to estimate the cost of debt.

Company	S&P Rating	Risk-Free Rate	Default Spread	Cost of Debt	Tax Rate	After-Tax Cost of Debt
Disney	A	2.75% (US \$)	1.00%	3.75%	36.1%	2.40%
Deutsche Bank	A	1.75% (€)	1.00%	2.75%	29.48%	1.94%
Vale	A-	2.75% (US \$)	1.30%	4.05%	34%	2.67%

- For Tata Motors, we have a rating of AA- from CRISIL, an Indian bond-rating firm, that measures only company risk. Using that rating:

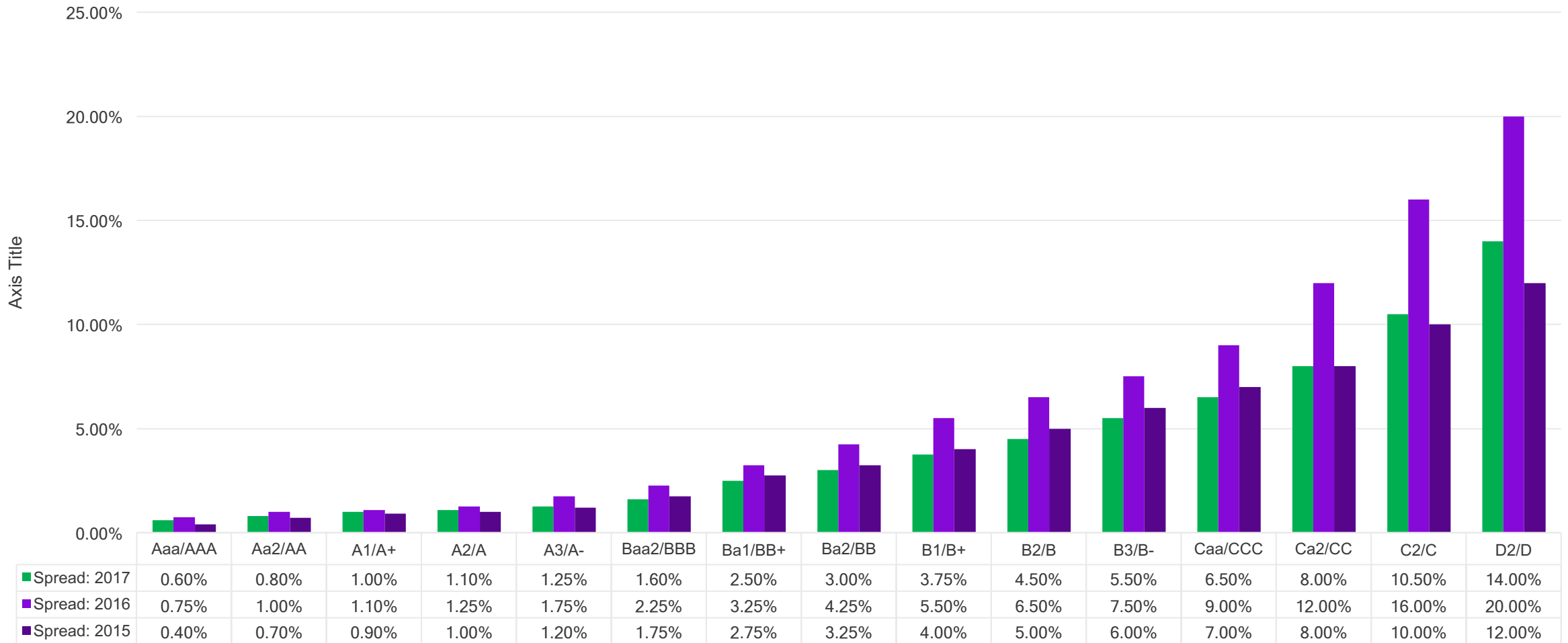
$$\text{COST OF DEBT}_{\text{TMT}} = \text{RISK-FREE RATE}_{\text{Rs}} + \text{DEFAULT SPREAD}_{\text{INDIA}} + \text{DEFAULT SPREAD}_{\text{TMT}}$$

$$= 6.57\% + 2.25\% + 0.70\% = 9.62\%$$

$$\text{AFTER-TAX COST OF DEBT} = 9.62\% (1-0.3245) = 6.50\%$$

UPDATED DEFAULT SPREADS – JANUARY 2017

Default Spreads for 10-year Corporate Bonds: 2015 thru 2017

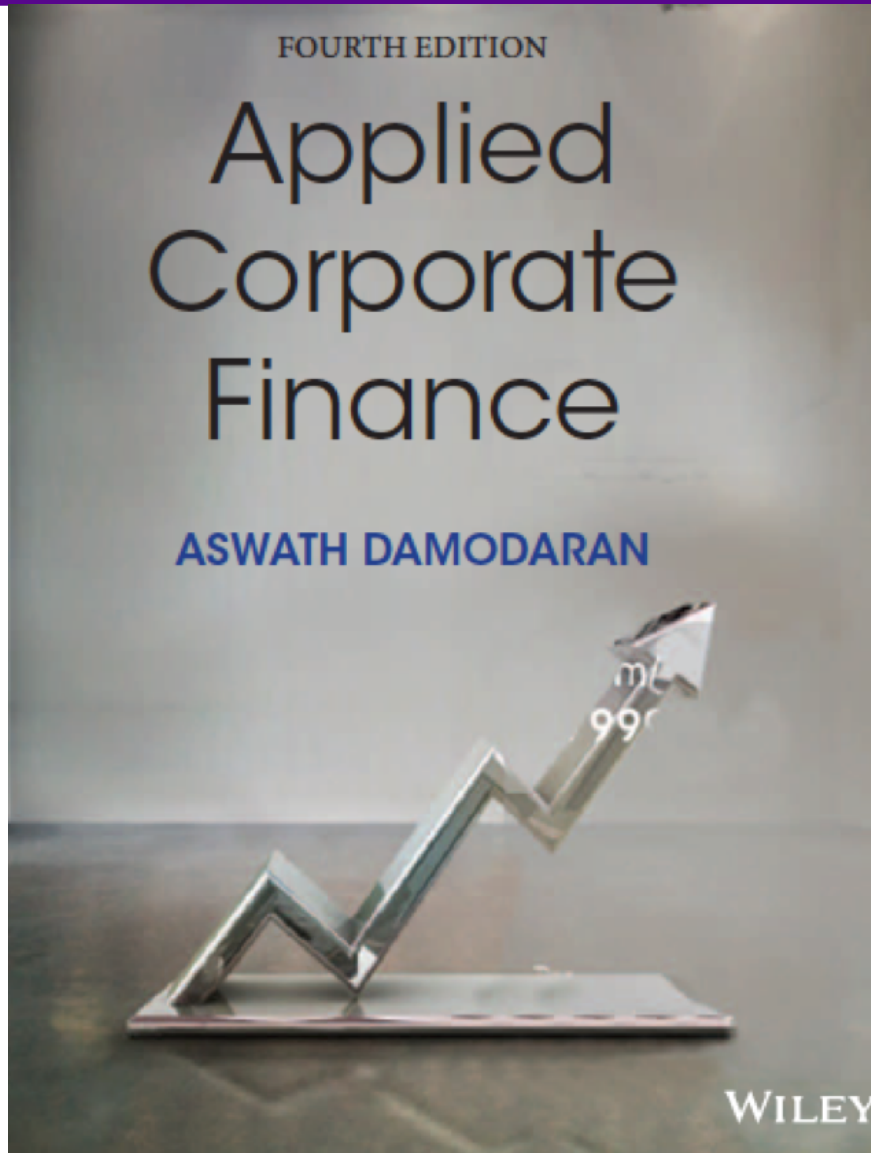




APPLICATION TEST: ESTIMATING A COST OF DEBT

- Based upon your firm's current earnings before interest and taxes, its interest and taxes, its interest expenses, estimate
 - An interest coverage ration for your firm
 - A synthetic rating for your firm (use the tables from prior pages)
 - A pre-tax cost of debt for your firm
 - An after-tax cost of debt for your firm

Task
Estimate
the cost of
debt for
your
company



Optional:
Read
Chapter 4